Probability · 4-3, 4-4 & 4-5 · One-Page Sheet

4-3 · INDEPENDENT VS DEPENDENT

- **Independent**: A occurring does not change P(B). **Dependent**: it does.
- **With replacement**, separate devices (coin + die), people from a large population → independent.
- **Without replacement**, "of those who...", "of these..." → dependent / conditional.
- Independent ≠ mutually exclusive. ME events with P > 0 are always **dependent**.

المستقلان: وقوع أحدهما لا يغيّر احتمال الآخر. السحب مع الإرجاع يعطي أحداثًا مستقلة، والسحب بدون إرجاع يعطي أحداثًا تابعة.

4-3 · MULTIPLICATION RULES ("AND")

Rule 1 (independent):

$$P(A \cap B) = P(A) \cdot P(B)$$

Rule 2 (dependent):

$$P(A \cap B) = P(A) \cdot P(B \mid A) = P(B) \cdot P(A \mid B)$$

n independent trials:

$$P(\text{all}) = p \cdot p \cdots p = p^n$$

Without replacement, chain it: 2 aces = (4/52)(3/51) = 1/221; each next fraction drops by 1 on top (same kind) and bottom.

4-3 · CONDITIONAL PROBABILITY

$$P(B \mid A) = P(A \cap B) / P(A) \quad P(A) \neq 0$$

The condition goes in the denominator.

"Given that", "if", "of those who", or a fact stated as already true ("he has to park in a no-parking zone").

Independence tests (any one): $P(B \mid A) = P(B)$ · $P(A \mid B) = P(A)$ · $P(A \cap B) = P(A)P(B)$

الشرط دائمًا في المقام. والاستقلال يعني أن الاحتمال الشرطي يساوي الاحتمال العادي.

4-3 · TWO-WAY TABLES

P(A and B)	cell ÷ **grand total**
P(A | B)	cell ÷ **total of B's row or column**
independent?	compare P(A | B) with P(A)
ME?	is the shared cell 0?

P(yes | female) ≠ P(female | yes): different denominators.

4-3 · TREE DIAGRAMS

Multiply along a branch (P of a path) · **add across** the paths that give the event · branches from each point sum to 1 · all end paths sum to 1.

$$P(B) = P(A) \cdot P(B \mid A) + P(A^c) \cdot P(B \mid A^c)$$

Reverse question:

$$P(A \mid B) = P(A \cap B) / P(B)$$

4-3 · "AT LEAST ONE"

$$P(\text{at least one}) = 1 - P(\text{none})$$

n independent tries, success p: $1 - (1 - p)^n$. Die 3 rolls, at least one 5: $1 - (5/6)^3 = 91/216$.

4-4 · FUNDAMENTAL COUNTING RULE

$$k_1 \times k_2 \times \cdots \times k_n$$

- Repetition **allowed**: same count each slot ($10 \times 10 \times 10 \times 10$).
- **Not allowed**: drop by 1 each slot ($10 \times 9 \times 8 \times 7$).
- Restricted slot first (first digit ≠ 0: $9 \times 9 \times 8 \times 7$).
- **and** → × · **or** → + (cases)

4-4 · FACTORIAL

$n! = n(n-1)\cdots 2 \cdot 1$ $0! = 1$ n! = ways to arrange n different objects in a row.

N	0	1	2	3	4	5	6	7	8
n!	1	1	2	6	24	120	720	5040	40320

Cancel: 8!/5! = 8·7·6 = 336.

4-4 · PERMUTATIONS (ORDER MATTERS)

Rule 1:

$$_nP_r = n! / (n - r)!$$

$_nP_n = n!$

Rule 2 (identical objects):

$$n! / (r_1!\, r_2! \cdots r_p!)$$

STATISTICS: 10!/(3! 3! 2!) = 50,400 · LEVEL: 5!/(2! 2!) = 30. Tally the letters first; the counts must sum to n.

4-4 · COMBINATIONS (ORDER DOESN'T MATTER)

$$_nC_r = n! / [(n - r)! \cdot r!] = {}_nP_r / r!$$

$_nC_r = {}_nC_{n-r}$ · $_nC_0 = {}_nC_n = 1$ · $_nC_1 = n$ · C ≤ P always.

Two groups: 3 of 7 women **and** 2 of 5 men = $_7C_3 \times {}_5C_2 = 35 \times 10 = 350$.

Must include a person: fix them, then $_{n-1}C_{r-1}$. Must exclude: $_{n-1}C_r$.

4-4 · P OR C? (DOES ORDER CHANGE THE RESULT?)

Permutation	rank, arrange, seat, line up, 1st/2nd/3rd, distinct offices, codes, PINs, plates, order of appearance
Combination	choose, select, committee or team (no titles), group, sample, hand of cards, lottery numbers

A "combination lock" is really a permutation. Casio: `n SHIFT × r =` nPr · `n SHIFT ÷ r =` nCr · `SHIFT x⁻¹` x!

إذا تغيّر الناتج بتغيير الترتيب فهو تباديل، وإلا فهو توافيق. وكلمة «و» تعني الضرب، وكلمة «أو» تعني الجمع.

4-5 · PROBABILITY WITH COUNTING RULES

1. **n(S)**: all ways to select (usually $_nC_r$).
2. **n(E)**: ways that meet the condition (product of C's, one per group).
3. `P(E) = n(E) / n(S)`, then a sentence.

$$P(k \text{ from group a, } r-k \text{ from group b}) = {}_aC_k \cdot {}_\beta C_{r-k} / {}_{a+\beta}C_r$$

Defective: 24 circuits, 4 bad, pick 4. P(exactly 2 bad) = $_4C_2 \cdot {}_{20}C_2 / {}_{24}C_4$ = 1140/10626 ≈ 0.1073. P(at least 1 bad) = 1 − $_{20}C_4 / {}_{24}C_4$ ≈ 0.5440.

Count n(E) and n(S) with the **same** rule. The C-method equals the without-replacement chain: $_9C_3 / {}_{12}C_3$ = (9/12)(8/11)(7/10).

احسب فضاء العينة ثم الحدث بنفس طريقة العدّ، ثم اقسم. «على الأقل واحد» = 1 − الكل من المجموعة الأخرى.

TRAPS THAT COST MARKS

1. Missing "without replacement": the second denominator must drop.
2. Putting P(B) under P(A ∩ B) when the question asks for P(B | A).
3. Treating "of those who..." as an unconditional probability.
4. Calling ME events independent.
5. Counting only one order in "one of each" (×2 in the chain method).
6. Permutation for a committee, or a combination for officers.
7. Adding instead of multiplying across groups.
8. 0! = 0 (it is 1); forgetting the repeated letters in rule 2.
9. A permutation numerator over a combination denominator.

WRITE IT LIKE A MARKER READS

1. Define events; say "with" or "without replacement".
2. Name the rule: "dependent, so Rule 2"; "order doesn't matter, so $_nC_r$".
3. Show n(E) and n(S) separately, then divide.
4. Simplify, give a decimal, check 0 ≤ P ≤ 1.